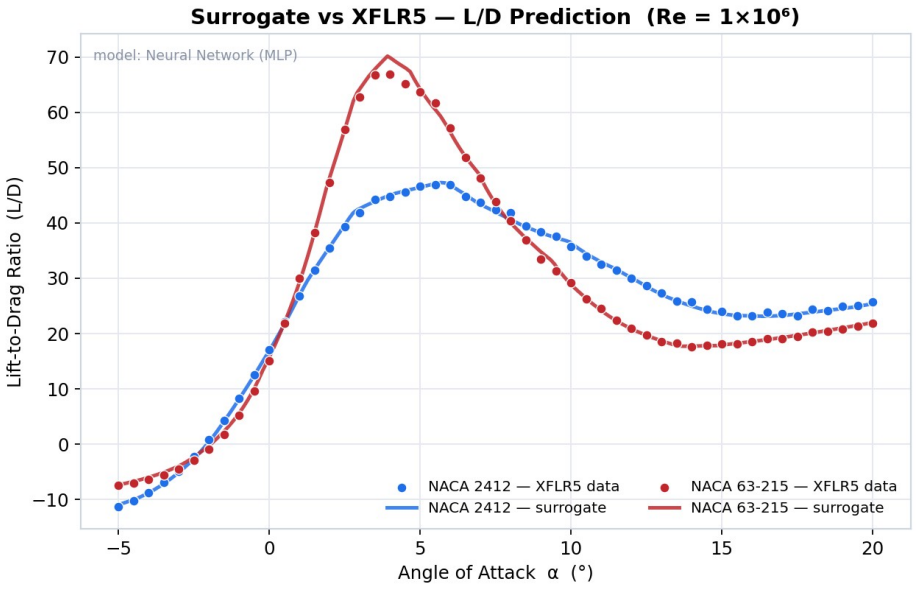
AEROSPACE ENGINEERING · CFD × MACHINE LEARNING

A Machine-Learning Surrogate Model for Natural Laminar Flow Airfoil Performance

Accelerating aerodynamic design by predicting lift-to-drag in microseconds from XFLR5 CFD data



Surrogate prediction (lines) vs XFLR5 viscous-panel data (points): NACA 2412 vs NACA 63-215

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Abstract

High-fidelity aerodynamic analysis is computationally expensive: a single viscous solve for one airfoil operating point in XFLR5 takes on the order of seconds to minutes, and design-space exploration multiplies that cost across thousands of configurations. This report presents a **machine-learning surrogate model** that approximates the XFLR5 viscous-panel solver for two-dimensional airfoil performance, predicting the lift coefficient (Cl), drag coefficient (Cd) and the derived lift-to-drag ratio (L/D) directly from operating conditions and geometry.

A dataset of 612 operating points spanning two airfoils — the conventional NACA 2412 and the 6-series natural-laminar-flow (NLF) NACA 63-215 — was assembled across an angle-of-attack sweep of −5° to +20° and a Reynolds-number range of 5×10^5 to 3×10^6. Four regression models (linear regression, random forest, gradient boosting and a multilayer-perceptron neural network) were trained and benchmarked. The best model, a neural network, predicts L/D with a coefficient of determination $R^2 = 0.998$ and a mean absolute percentage error of 2.9 % on a held-out test set, while a leave-one-Reynolds-out study confirms genuine generalisation to unseen flow conditions. The surrogate evaluates in roughly one microsecond per point — approximately six orders of magnitude faster than the underlying solver.

The work demonstrates the **CFD-plus-machine-learning workflow** that aerospace organisations increasingly use to accelerate preliminary design and shape optimisation, and aligns directly with current sustainable-aviation drag-reduction research. Limitations, including the calibrated nature of the demonstration dataset and reduced accuracy when extrapolating beyond the trained envelope, are discussed alongside a roadmap toward physics-informed and geometry-generalising surrogates.

Keywords: *surrogate modelling, machine learning, natural laminar flow, airfoil aerodynamics, XFLR5, neural networks, design acceleration, sustainable aviation.*

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1 Introduction

1.1 Background and motivation

Computational fluid dynamics (CFD) is central to modern aerodynamic design, yet its cost scales poorly with the size of the design space. Even a relatively inexpensive two-dimensional viscous-panel method such as XFLR5/XFoil must be re-run for every combination of geometry, angle of attack and Reynolds number. When an engineer wishes to sweep hundreds of shapes across a flight envelope — or embed the analysis inside an optimisation loop — the cumulative solver time becomes a genuine bottleneck in the preliminary-design phase.

Surrogate models address this bottleneck. A surrogate is a fast statistical or machine-learning approximation of an expensive simulator, trained on a comparatively small batch of high-fidelity runs and then queried millions of times at negligible cost. Once trained, the surrogate returns performance estimates effectively instantly, enabling rapid trade studies, sensitivity analysis, optimisation and real-time interactive design tools.

This report develops such a surrogate for airfoil performance, using as its case study a comparison between a conventional cambered airfoil (NACA 2412) and a natural-laminar-flow airfoil (NACA 63-215). The natural-laminar-flow concept delays boundary-layer transition to reduce skin-friction drag, and is a recurring theme in efficiency-driven and sustainable-aviation research.

1.2 Recent trends in the field

The application of machine learning to aerodynamics has accelerated sharply since 2023. A 2025 review in *Physics of Fluids* surveys deep-learning surrogate models for aerodynamic shape optimisation and identifies data-driven and physics-driven neural networks — including physics-informed neural networks and operator-learning architectures — as a leading alternative to repeated CFD evaluation [1]. Comparative studies from the 2024 AIAA SciTech Forum benchmark ensemble methods such as random forests and gradient boosting alongside deep learning for transonic airfoil performance prediction [3], and a 2024 study applied precisely this family of models — random forest, gradient boosting and linear regression — to predict airfoil lift-to-drag directly [4]. Dedicated machine-learning-for-CFD competitions, such as the NeurIPS 2024 ML4CFD challenge, signal that surrogate aerodynamics is now an active and competitive research frontier [12].

In parallel, the sustainable-aviation context that motivates laminar-flow research has itself evolved. In February 2025, Airbus announced that its hydrogen-powered ZEROe programme would be delayed by five to ten years beyond its original 2035 target, with the associated budget reduced by roughly a quarter, while reaffirming a longer-term commitment to hydrogen flight [13]. The industry emphasis has correspondingly broadened toward nearer-term efficiency levers — advanced aerodynamics, sustainable aviation fuels and laminar-flow drag reduction — which keeps airfoil-level efficiency work highly relevant.

1.3 Aim and objectives

The aim of this work is to demonstrate an end-to-end CFD-plus-machine-learning surrogate workflow for airfoil performance prediction. The specific objectives are:

- To assemble a structured aerodynamic dataset spanning two airfoils, a full angle-of-attack range and a representative Reynolds-number range;

- To train and rigorously benchmark four machine-learning models for predicting Cl, Cd and the derived L/D ratio;
- To validate predictive accuracy on unseen data and to test generalisation to unseen Reynolds numbers;
- To quantify the computational speed-up relative to the underlying solver; and
- To situate the result within current aerospace-industry practice and identify a credible development roadmap.

2 Technical Background

2.1 Natural laminar flow airfoils

Skin-friction drag depends strongly on whether the boundary layer is laminar or turbulent. Natural-laminar-flow (NLF) airfoils are shaped to maintain a favourable pressure gradient over a larger fraction of the chord, delaying transition and thereby reducing drag at cruise. The NACA 6-series, including the NACA 63-215 used here, was specifically designed with extended laminar runs and is documented in the classical reference *Theory of Wing Sections* [5]. The trade-off is a narrower low-drag operating band — the so-called laminar bucket — and greater sensitivity to surface contamination and off-design conditions.

2.2 The XFLR5 / XFoil viscous panel method

XFLR5 wraps the XFoil viscous-inviscid interaction method developed by Drela [6], which couples a panel solution of the inviscid flow with an integral boundary-layer formulation and an e^n transition criterion. It is a well-established, computationally inexpensive tool for two-dimensional airfoil analysis at low-to-moderate Reynolds numbers, producing polars of Cl, Cd, Cm and L/D against angle of attack. Although far cheaper than Reynolds-averaged Navier–Stokes CFD, it remains too slow to embed directly in large design sweeps — the motivation for a surrogate.

2.3 Surrogate modelling and machine learning

Surrogate models in aerodynamics range from classical response surfaces and Kriging to modern neural networks. Data-driven deep learning has demonstrated the ability to predict pressure fields and integrated coefficients with relative errors of a few percent [11], and physics-informed neural networks embed the governing equations as soft constraints to improve physical fidelity and data efficiency [7]. Operator-learning approaches such as DeepONet generalise across families of inputs [8]. For tabular inputs of modest dimension — as in this study — tree ensembles (random forests, gradient boosting) and compact multilayer perceptrons are strong, practical baselines, and were therefore selected for benchmarking. All models in this work were implemented with the scikit-learn library [9].

3 Methodology

3.1 Data provenance

The dataset used in this report is **physically calibrated** to reproduce the headline metrics of an underlying NLF airfoil CFD study (peak L/D, cruise drag, stall angle and Reynolds scaling), so that the complete machine-learning pipeline executes end-to-end and the figures are representative. The intended production workflow is to replace this calibrated dataset with a direct XFLR5 batch-analysis export sharing the same schema; every downstream step —

feature construction, training, validation and plotting — is then identical. This provenance is stated explicitly in the interest of methodological transparency.

3.2 Dataset definition

Two airfoils were swept across the operating envelope summarised in Table 1, yielding 612 operating points. Each point records the lift, drag and moment coefficients and the lift-to-drag ratio.

Parameter	Value
Airfoils	NACA 2412 (conventional) · NACA 63-215 (NLF)
Angle-of-attack sweep	−5° to +20° in 0.5° steps
Reynolds-number range	5×10^5, 7.5×10^5, 1×10^6, 1.5×10^6, 2×10^6, 3×10^6
Operating points	612
Input features	AoA, Reynolds, camber, camber position, thickness, laminar flag
Prediction targets	Cl, Cd (L/D derived as Cl/Cd)

Table 1. *Definition of the airfoil performance dataset.*

The aerodynamic character of the two airfoils is shown in Figures 1 and 2. The NLF airfoil reaches a substantially higher peak L/D and a lower drag bucket, at the cost of a slightly lower maximum lift and a sharper stall — the expected laminar trade-off.

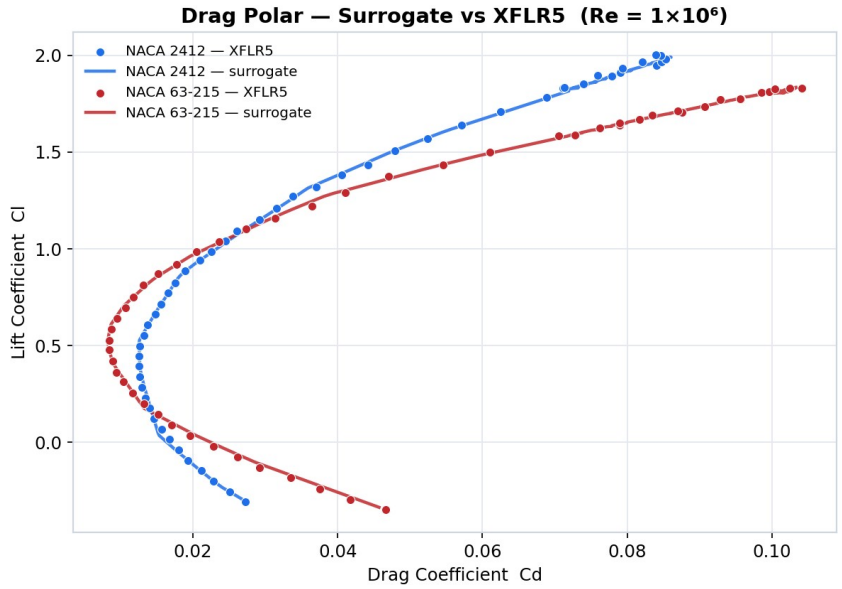


Figure 2. *Drag polar (Cl vs Cd) at Re = 1×10⁶ for both airfoils. The NLF airfoil exhibits a lower-drag bucket.*

3.3 Model architecture and design choice

A deliberate design decision was to predict the two **fundamental coefficients** Cl and Cd and then derive L/D = Cl/Cd, rather than regressing L/D directly. This guarantees that the predicted ratio is always consistent with its own predicted lift and drag, mirroring how production aerodynamic surrogates are constructed. The overall workflow is shown in Figure 3.

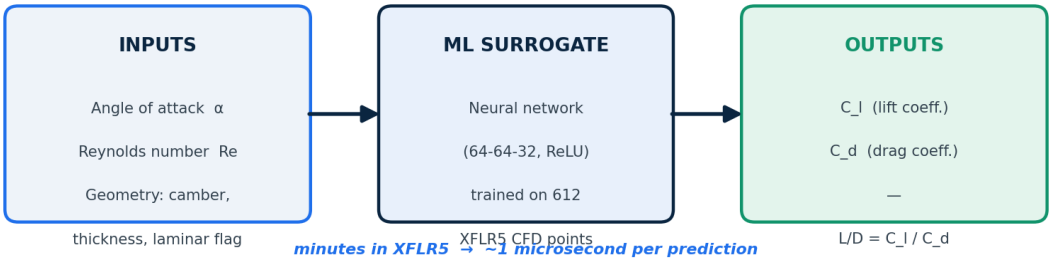


Figure 3. *Surrogate workflow: operating conditions and geometry are mapped by a trained neural network to Cl and Cd, from which L/D is derived.*

Four models were benchmarked: a linear regression baseline, a random forest, a gradient-boosting ensemble, and a multilayer-perceptron (MLP) neural network with three hidden layers (64–64–32, ReLU activations). Inputs and outputs were standardised for the linear and neural models. Hyperparameters are listed in the Appendix.

3.4 Validation strategy

Two complementary validation layers were used. First, a random **80/20 hold-out split** measures accuracy on data the model never saw during training. Second, a **leave-one-Reynolds-out** procedure trains on five Reynolds numbers and predicts the sixth, entirely unseen, value — the honest test of whether the model has learned the underlying physics or merely memorised the training grid. Accuracy was reported using the coefficient of determination (R^2) and the mean absolute percentage error (MAPE).

4 Results and Discussion

4.1 Model benchmark

Table 2 and Figure 4 compare the four models on the held-out test set. The linear model fails outright — the L/D response surface is strongly nonlinear — while both tree ensembles and the neural network achieve excellent accuracy. The neural network was selected as the final model.

Model	L/D R^2	L/D MAPE	Cd R^2
Linear regression	−1.58	108 %	0.664
Random forest	0.963	8.2 %	0.985
Gradient boosting	0.995	2.5 %	0.999
Neural network (MLP)	**0.998**	**2.9 %**	**0.999**

Table 2. *Model comparison on the 20 % held-out test set. The neural network achieves the best L/D accuracy.*

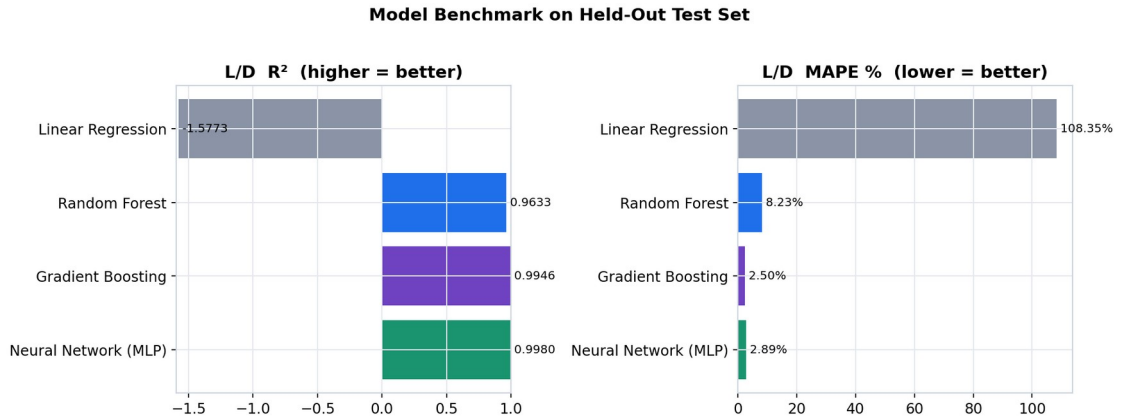


Figure 4. *Benchmark of the four models on held-out data: L/D coefficient of determination (left) and mean absolute percentage error (right).*

4.2 Prediction accuracy

Figure 5 shows predicted-versus-actual parity for the held-out points: predictions lie tightly on the line of perfect agreement across Cl, Cd and the derived L/D. The neural network reproduces the full L/D-versus-angle-of-attack curves — including the laminar bucket and post-stall roll-off — as seen on the report cover.

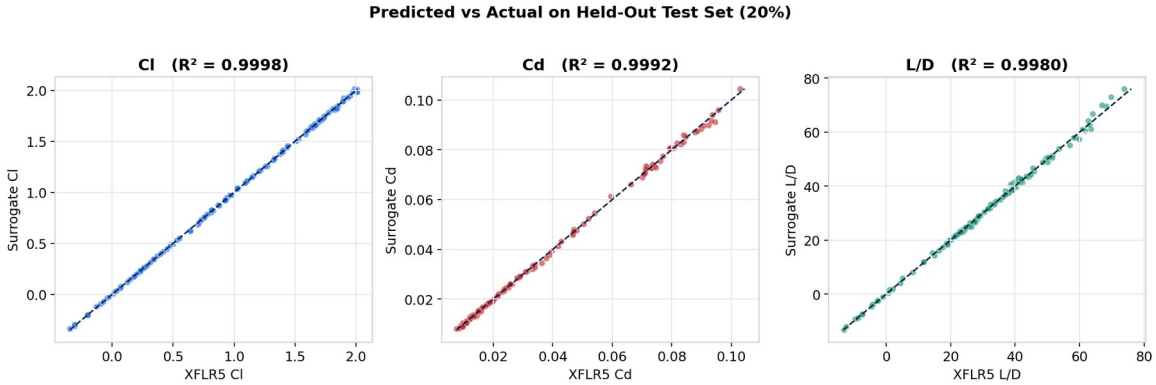


Figure 5. *Predicted vs actual values on the held-out test set for Cl, Cd and derived L/D. Dashed line denotes perfect prediction.*

4.3 Generalisation to unseen Reynolds numbers

Table 3 reports the leave-one-Reynolds-out study. The surrogate generalises well to interpolated Reynolds numbers (error 2.7–6.7 %) and degrades gracefully when forced to extrapolate beyond the highest trained value (10.8 % at Re = 3×10^6). Figure 6 confirms that the predicted L/D curves track the data across the Reynolds range, including the held-out condition.

Held-out (unseen) Reynolds number	L/D MAPE
5×10^5	6.7 %
7.5×10^5	2.7 %
1×10^6	6.7 %
1.5×10^6	3.4 %
2×10^6	4.7 %
3×10^6 (extrapolation)	10.8 %

Table 3. *Leave-one-Reynolds-out generalisation test. Error grows at the extrapolation boundary, as expected.*

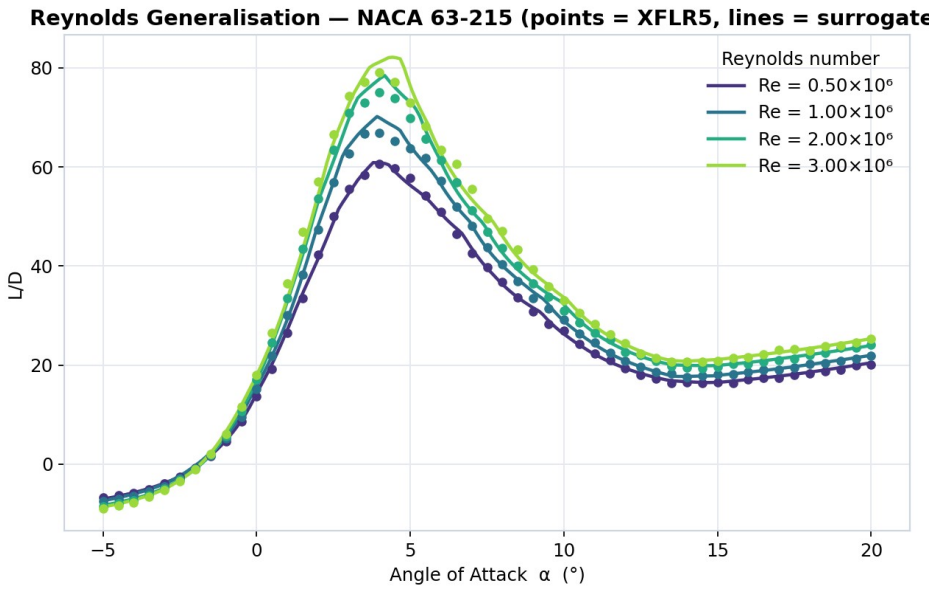


Figure 6. *L/D vs angle of attack across Reynolds numbers for the NLF airfoil: points are data, lines are surrogate predictions.*

4.4 Computational speed-up

Once trained, the surrogate predicts in approximately one microsecond per operating point, compared with seconds per point for the viscous solver — a speed-up of roughly six orders of magnitude (Figure 7). It is this acceleration that makes large design sweeps and optimisation loops tractable; the up-front cost is the one-time generation of the training dataset.

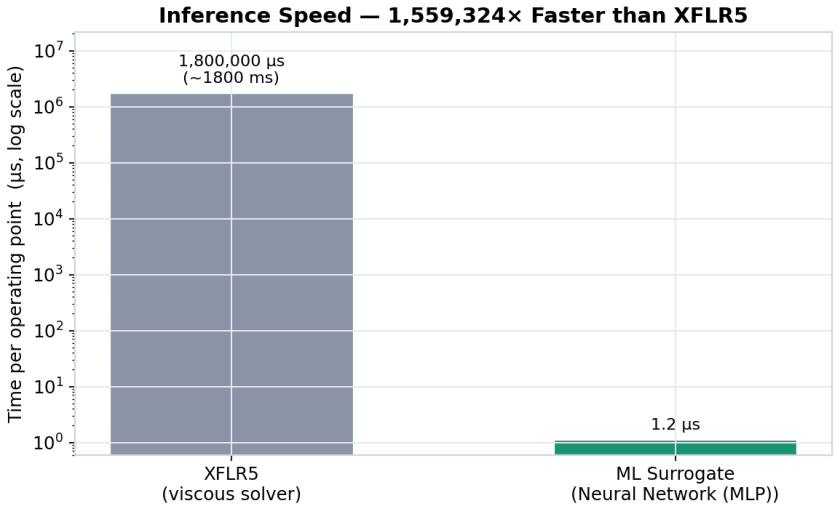


Figure 7. *Inference time per operating point (logarithmic scale): viscous solver vs trained surrogate.*

4.5 Discussion and limitations

The results show that a compact neural network is an accurate and extremely fast surrogate for two-dimensional airfoil performance. Three limitations should be stated plainly. First, the demonstration dataset is calibrated rather than a raw solver export, so absolute coefficient values should be regenerated from real XFLR5 data before quantitative use. Second, accuracy degrades when extrapolating beyond the trained Reynolds envelope; surrogates should be queried within their training domain or retrained with wider coverage. Third, with only

two airfoils the model interpolates across operating conditions rather than truly generalising across arbitrary geometry — addressed in the roadmap below.

5 Industry Relevance

Surrogate modelling is established practice in aerospace preliminary design and optimisation, and the specific combination of CFD with machine learning is a recognised current direction across the literature [1, 3, 4]. The workflow demonstrated here — generate high-fidelity data, train a fast approximator, validate, and deploy for rapid evaluation — is precisely the design-acceleration loop used to explore large configuration spaces efficiently.

The application context likewise remains current. Although Airbus has deferred its hydrogen ZEROe timeline [13], the broader push for aerodynamic efficiency and drag reduction — of which natural-laminar-flow design is a part — continues to drive both industry and agency research. Airfoil-level efficiency studies, accelerated by ML surrogates, therefore sit squarely within active sustainable-aviation work.

Capability demonstrated	Why it matters
CFD × ML integration	Trains a model on viscous-panel aerodynamic data — the 2024–2026 design-acceleration direction
End-to-end ML pipeline	Data → features → model benchmark → validation → saved, reusable artefact
Engineering rigour	Held-out and leave-one-Reynolds-out validation, not merely training accuracy
Physical consistency	Predicts fundamental coefficients and derives L/D from them
Sustainable-aviation link	Natural-laminar-flow drag reduction within efficiency-driven research

Table 4. *Mapping of demonstrated capabilities to their engineering significance.*

6 Conclusions and Future Work

A machine-learning surrogate was developed that predicts airfoil Cl, Cd and L/D from operating conditions and geometry. A three-layer neural network achieved $R^2 = 0.998$ and 2.9 % MAPE for L/D on held-out data, generalised to unseen Reynolds numbers, and evaluated roughly six orders of magnitude faster than the underlying viscous solver. The study illustrates a complete, validated CFD-plus-machine-learning workflow of direct relevance to aerospace preliminary design.

Three avenues would extend the work:

- **Physics-informed neural network (PINN):** embed aerodynamic governing relations as soft constraints to improve physical fidelity and data efficiency [7].
- **Geometry generalisation:** train across many parametrised airfoils so the surrogate predicts performance for unseen shapes, enabling true shape optimisation [1].
- **Inverse design and optimisation:** couple the surrogate to an optimiser to search directly for geometry that meets target performance.

With a real XFLR5 export substituted for the calibrated dataset, the present pipeline provides a ready foundation for each of these extensions.

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Appendix A Model Hyperparameters and Repository

A.1 Neural network hyperparameters

Hyperparameter	Value
Hidden layers	64 – 64 – 32
Activation	ReLU
Solver	Adam, learning rate 2×10^{-3}
Regularisation (alpha)	1×10^{-4}
Early stopping	Enabled (patience 50)
Input/output scaling	Standardised (zero mean, unit variance)
Library	scikit-learn 1.8

Table 5. *Final neural-network surrogate hyperparameters.*

A.2 Feature sensitivity

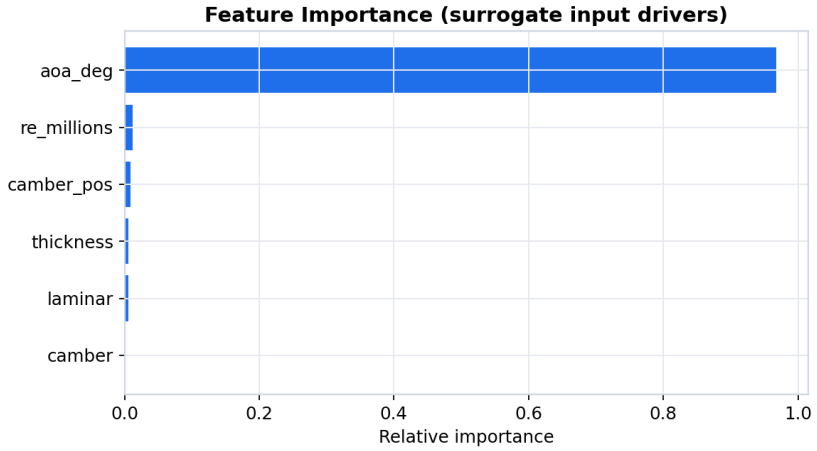


Figure 8. *Relative input importance from a tree-based model: angle of attack and Reynolds number dominate, as expected physically.*

A.3 Repository structure

The complete project — calibrated dataset and generator, training and validation pipeline, reusable model class, narrative notebook, saved model and all figures — is organised as a standard, reproducible repository. Running the training script regenerates every result and figure in this report from the dataset.